

Appendices

APPENDIX A

Review of Integration Techniques

In solving differential equations, we cannot overemphasize the importance of integration techniques. Indeed, we sometimes refer to the problem of solving a differential equation by asking, “How can we integrate this equation?”

This section is devoted to reviewing three of the standard techniques that will be useful for integrating many of the functions encountered in this text. Of course, one can always search in a table of integrals (a brief table appears on the inside back cover of this text), but familiarity with tricks of integration is a worthwhile investment of time for several reasons: (i) we may not find the particular integral we need in the table; (ii) using one of the techniques may be quicker; and (iii) we might be curious to know how the various integrals in the table were obtained.

So what techniques can we use for the following integrals?

$$\begin{array}{lll} \text{(a)} \int x\sqrt{2x+1} \, dx & \text{(b)} \int \frac{x^2}{\sqrt{9-x^2}} \, dx & \text{(c)} \int x \ln x \, dx \\ \text{(d)} \int \frac{7x^2 + 10x - 1}{x^3 + 3x^2 - x - 3} \, dx & \text{(e)} \int e^{x^2} \, dx & \end{array}$$

For **(a)** we can use *algebraic substitution*, while **(b)** can be handled with *trigonometric substitution*. For **(c)** we have available *integration by parts*, and the integral in **(d)** will yield to *partial fractions*. But try as we may, no trick is sufficient to give **(e)** in a finite number of terms. By this we mean that if we have available the elementary functions (rational functions, trigonometric functions, inverse trigonometric functions, exponential functions, and logarithmic functions) and permit use of the standard operations—addition, subtraction, multiplication, division, taking roots, and composite functions—a finite number of times, we cannot find a formula for the antiderivative of e^{x^2} . A similar conundrum arises with integration of $\sqrt{1 + \cos^2 x}$. So we can add a fourth reason for honing our skills with integration techniques: They will help us to recognize those integrals that *can* be expressed in a finite number of terms.

In this context, we are speaking of integration in closed form—analytically—as opposed to numerical integration. Thus integration is the inverse of differentiation; it requires finding an *antiderivative*. As a warmup, then, we begin this appendix by jogging your recollection of rules for *derivatives*. We hope you will find our approach a little different and refreshing.

Differentiation

Of course you remember that the derivative of a constant equals zero, and the derivative of a sum equals the sum of the derivatives. So let’s jump in with the basic power rule.

Derivative of a power function $\frac{d}{dx}x^p = px^{p-1}.$

This holds for *any* constant exponent p , be it integer, fraction, or negative.

Examples: $\frac{d}{dx}x^{100} = 100x^{99}$, $\frac{d}{dx}x^{7/2} = \frac{7}{2}x^{5/2}$, $\frac{d}{dx}x^{1/3} = \frac{1}{3}x^{-2/3}$, $\frac{d}{dx}x^{-e} = -ex^{-e-1}$

Derivative of the exponential $\frac{d}{dx}e^x = e^x$.

This is the essential feature of the exponential function.

Derivative of trig functions $\frac{d}{dx}\sin x = \cos x$, $\frac{d}{dx}\cos x = -\sin x$.

Remember which one gets the minus sign.

Derivative of a product $\frac{d}{dx}(fg) = f\frac{d}{dx}g + g\frac{d}{dx}f$.

Example: $\frac{d}{dx}(x^2 \sin x) = x^2 \cos x + (\sin x)(2x)$

(Bonus: If we rewrite this rule as $f\frac{d}{dx}g = \frac{d}{dx}(fg) - g\frac{d}{dx}f$, we begin to see how to build an antiderivative (fg) for $f\frac{d}{dx}g$, by compensating with $-g\frac{d}{dx}f$. This is the essence of *integration by parts*.)

Derivative of a quotient $\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{g\frac{d}{dx}f - f\frac{d}{dx}g}{g^2}$.

(If you forget this, you can just apply the product rule to fg^{-1})

Examples:

$$\begin{aligned}\frac{d}{dx}\tan x &= \frac{\frac{d}{dx}\sin x}{\cos x} = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} = \sec^2 x = 1 + \tan^2 x\end{aligned}$$

(Bonus: if you reverse the roles, so $\tan x = y$ and $x = \arctan y$, and invert, you get a new rule:

$$\frac{d}{dy}\arctan y = \frac{1}{1+y^2})$$

$$\frac{d}{dx}\sec x = \frac{\frac{d}{dx}1}{\cos x} = \frac{(\cos x)(0) - 1(-\sin x)}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = (\sec x)(\tan x)$$

Chain rule $\frac{d}{dx}f(u) = \left[\frac{d}{du}f(u)\right]\left[\frac{du}{dx}\right] \cdot \left(\text{Think } \frac{df(u)}{du} \frac{du}{dx}\right)$

Examples: $\frac{d}{dx}(7x^2 + x - 2)^{-1/4} = \left[-\frac{1}{4}(7x^2 + x - 2)^{-5/4}\right]\left[\frac{d}{dx}(7x^2 + x - 2)\right]$
 $= -\frac{1}{4}\frac{14x + 1}{(7x^2 + x - 2)^{5/4}}$

$$\frac{d}{dx}e^{x^2} = [e^{(x^2)}]\left[\frac{d}{dx}(x^2)\right] = 2xe^{x^2}.$$

Now we begin our review of integration with the *method of substitution* (also called *change of variable*), which is derived from the chain rule.

Method of Substitution

From the formula $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$ we obtain

$$\int f'(g(x))g'(x)dx = f(g(x)) + C.$$

Thus if we can recognize an integrand as being of the form $f'(g(x))g'(x)$, then integration is immediate via the substitution $u = g(x)$. Here are some simple illustrations:

- $\int \sin^3 2x \cos 2x dx$

Since $d(\sin 2x) = 2 \cos 2x dx$, the substitution $u = \sin 2x$ reduces the integration to that of $u^3/2 du$.

- $\int x^3 e^{5x^4} dx$

Observe that x^3 is essentially (except for a constant factor) the derivative of $5x^4$, so we set $u = 5x^4$ and integrate $e^u/20 du$.

- $\int \frac{6 \ln(x+5)}{x+5} dx$

Recognizing that $1/(x+5)$ is the derivative of $\ln(x+5)$, we put $u = \ln(x+5)$ and integrate $6u du$.

- $\int \tan^3 x \sec x dx$

Here it is convenient to use the identity $\tan^2 x = \sec^2 x - 1$ and write the integrand as $(\sec^2 x - 1)\sec x \tan x$. Recalling that $(\sec x)' = \sec x \tan x$, the substitution $u = \sec x$ reduces the problem to integrating $(u^2 - 1) du$.

When a portion of the integrand involves some fractional power of an expression, an algebraic substitution can sometimes simplify the integration process. The technique is amply illustrated in the following example.

Example 1 Find

$$(1) \quad I_1 := \int x\sqrt{2x+1} dx.$$

Solution The difficulty here arises from the square root factor, so let's try to rid ourselves of this problem term by making the substitution $u = \sqrt{2x+1}$. Here are the details:

$$(2) \quad u^2 = 2x+1, \quad x = \frac{u^2-1}{2}, \quad dx = u du.$$

If we use these expressions in (1), we find that in terms of the new variable u ,

$$\begin{aligned}
 (3) \quad \int x\sqrt{2x+1} \, dx &= \int \left(\frac{u^2-1}{2} \right) (u) (u \, du) = \frac{1}{2} \int (u^4 - u^2) \, du \\
 &= \frac{1}{2} \left(\frac{u^5}{5} - \frac{u^3}{3} \right) + C \\
 &= \frac{u^3}{30} (3u^2 - 5) + C.
 \end{aligned}$$

Returning to the original variable x , via $u = \sqrt{2x+1}$, we deduce that

$$I_1 = \frac{(2x+1)^{3/2}}{30} [3(2x+1) - 5] + C = \frac{(2x+1)^{3/2}}{15} (3x-1) + C. \quad \blacklozenge$$

We remark that the substitution $u = \sqrt{2x+1}$ is not the only substitution that will allow us to compute the integral in (1). Another substitution that works just as well is $u = 2x+1$.

In a similar manner we can compute the integrals

$$\int \frac{x^3}{(1+x^2)^{3/2}} \, dx \quad \text{and} \quad \int \frac{e^{x^{1/3}+1}}{x^{2/3}} \, dx$$

by making the substitution $u = 1+x^2$ in the first and $u = x^{1/3}+1$ in the second.

The integration of certain algebraic expressions can sometimes be simplified by utilizing *trigonometric substitutions*. The effectiveness of these substitutions in handling quadratic expressions under a radical derives from the familiar identities

$$1 - \sin^2 \theta = \cos^2 \theta, \quad 1 + \tan^2 \theta = \sec^2 \theta, \quad \sec^2 \theta - 1 = \tan^2 \theta.$$

Accordingly, if the integrand involves:

- (I) $\sqrt{a^2 - x^2}$, set $x = a \sin \theta$,
- (II) $\sqrt{a^2 + x^2}$, set $x = a \tan \theta$,
- (III) $\sqrt{x^2 - a^2}$, set $x = a \sec \theta$.

These substitutions are easy to remember if we associate with each of the cases a right triangle whose sides are a , x , and a suitable radical.[†] The labeling of the triangle depends on the particular case at hand. For example, in (II) it is clear that the hypotenuse must be $\sqrt{a^2 + x^2}$, while in (I) the hypotenuse must be a , and in (III) the hypotenuse is x .

Example 2 Find the indefinite integral $I_2 := \int \frac{x^2 dx}{\sqrt{9-x^2}}$.

Solution This integral is of type (I) with $a = 3$. We first construct a triangle as shown in Figure A.1.

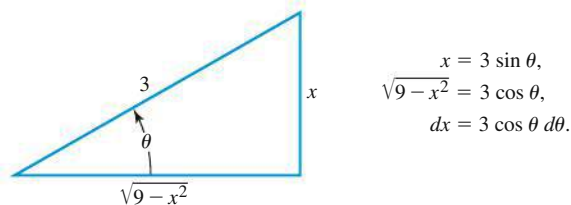


Figure A.1 Triangle for Example 2

[†]The radical sign may not be visible. For example, if the integrand is $1/(a^2 + x^2)^5$, this comes under case (II), because $1/(a^2 + x^2)^5 = 1/(\sqrt{a^2 + x^2})^{10}$.

From this triangle we see immediately that

$$I_2 = \int \frac{x^2 dx}{\sqrt{9-x^2}} = \int \frac{(3 \sin \theta)^2 3 \cos \theta d\theta}{3 \cos \theta} = 9 \int \sin^2 \theta d\theta .$$

To integrate $\sin^2 \theta$ we use the trigonometric identity

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

and obtain

$$\begin{aligned} I_2 &= \frac{9}{2} \int (1 - \cos 2\theta) d\theta = \frac{9}{2} \theta - \frac{9}{4} \sin 2\theta + C \\ &= \frac{9}{2} \theta - \frac{9}{2} \sin \theta \cos \theta + C . \end{aligned}$$

Returning to the original variable x , we can derive the expressions for θ , $\sin \theta$, and $\cos \theta$ directly from the triangle in Figure A.1. We find that

$$\theta = \arcsin \frac{x}{3}, \quad \sin \theta = \frac{x}{3}, \quad \cos \theta = \frac{\sqrt{9-x^2}}{3} .$$

Hence

$$I_2 = \frac{9}{2} \arcsin \frac{x}{3} - \frac{9}{2} \frac{x}{3} \frac{\sqrt{9-x^2}}{3} + C = \frac{9}{2} \arcsin \frac{x}{3} - \frac{1}{2} x \sqrt{9-x^2} + C . \quad \blacklozenge$$

Integration by Parts

The method of integration by parts is of both practical and theoretical importance. It is based on the product rule for differentiation which we now write in the format

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx} = uv' + vu' .$$

Integrating both sides of this equation and transposing, we obtain

$$(4) \quad \int uv' dx = uv - \int vu' dx .$$

Since $dv = v' dx$ and $du = u' dx$, equation (4) can be put in the more compact form

$$(5) \quad \int u dv = uv - \int v du .$$

Equation (5) is the basic equation for integration by parts. The idea is that the integrand on the left side may be very complicated, but if we select the part for u and the part for dv quite carefully, the integral on the right side may be much easier. (Some experimentation may be necessary to find the right selection.)

Example 3 Find the indefinite integral $I_3 := \int x \ln x \, dx$.

Solution There are two natural ways of selecting u and dv so that the product is $x \ln x \, dx$:

First possibility: let $u = x$, $dv = \ln x \, dx$.

Second possibility: let $u = \ln x$, $dv = x \, dx$.

With the first selection we cannot easily find v , the integral of $\ln x$, so we come to a dead end. In the second case we can find v . We arrange the work as follows. Let

$$u = \ln x, \quad dv = x \, dx.$$

Then

$$du = \frac{1}{x} \, dx, \quad v = \frac{x^2}{2}.$$

Making these substitutions in (5), we have for the left side

$$\int u \, dv = \int (\ln x) (x \, dx) = \int x \ln x \, dx.$$

and for the right side

$$(6) \quad uv - \int v \, du = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \frac{1}{x} \, dx = \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, dx.$$

Therefore, by (5),

$$(7) \quad I_3 = \int x \ln x \, dx = \frac{x^2}{2} \ln x - \frac{1}{4} x^2 + C. \quad \blacklozenge$$

Note that we used $v = x^2/2$ rather than $v = x^2/2 + C$. It is clear that equation (5) is correct for every value of C , and in most cases it is simpler to set $C = 0$. The arbitrary constant that we need can be added in the last step, as we have done in going from equation (6) to (7).

It is sometimes necessary to apply integration by parts more than once in order to evaluate a particular integral, such as

$$I_4 := \int x^2 e^x \, dx.$$

In the first application, we set $u = x^2$, $dv = e^x \, dx$ to deduce that

$$(8) \quad I_4 = x^2 e^x - 2 \int x e^x \, dx$$

and to evaluate the last integral in (8) we perform a second application of integration by parts with $u = x$, $dv = e^x \, dx$.

There are some integrals that may appear at first not to be split into two parts, but we need to keep in mind that “1” can be considered as a factor of any expression. For example, to compute the integrals

$$I_5 := \int \arctan(3x) \, dx \quad \text{and} \quad I_6 := \int \ln x \, dx.$$

an effective splitting for evaluating I_5 is $u = \arctan(3x)$, $dv = 1 \, dx$, while for I_6 an appropriate choice is $u = \ln x$, $dv = 1 \, dx$. The reader should verify that integration by parts then reduces these integrals to routine problems.

Partial Fractions

Any rational function, that is, any function of the form $R(x) = P(x)/Q(x)$, where $P(x)$ and $Q(x)$ are polynomials, can be integrated in finite terms. The first step in the process is to compare the degree of $P(x)$ with that of $Q(x)$. If $\deg P \geq \deg Q$, then division gives

$$(9) \quad \frac{P(x)}{Q(x)} = s(x) + \frac{p(x)}{Q(x)},$$

where $s(x)$ and $p(x)$ are polynomials and $\deg p < \deg Q$. Since the integration of the polynomial $s(x)$ is a trivial matter, we need only focus on the integration of the remainder term $p(x)/Q(x)$.

The idea of the method of partial fractions is to express $p(x)/Q(x)$ as the sum of simpler fractions that are easy to integrate. For example, knowing that

$$(10) \quad R(x) = \frac{7x^2 + 10x - 1}{x^3 + 3x^2 - x - 3}$$

can be written as

$$\frac{2}{x-1} + \frac{1}{x+1} + \frac{4}{x+3}$$

leads immediately to the evaluation

$$\int R(x) dx = 2 \ln|x-1| + \ln|x+1| + 4 \ln|x+3| + C.$$

The key to finding such a simple fraction representation is to first factor the denominator polynomial $Q(x)$. Indeed, in (10) the denominator is just the product $(x-1)(x+1)(x+3)$. In general, we recall that any polynomial $Q(x)$ with real coefficients can be factored into a product of linear and irreducible quadratic factors with real coefficients. There are four cases that may arise in the factorization: (i) nonrepeated linear factors; (ii) repeated linear factors; (iii) nonrepeated quadratic factors; and (iv) repeated quadratic factors.

In the case when a nonrepeated linear factor $(x-a)$ occurs in the factorization of $Q(x)$, we associate the partial fraction $A/(x-a)$, where A is a constant to be determined. If a linear factor $(x-a)$ is repeated m times in the factorization of $Q(x)$, we associate the sum of m fractions

$$\sum_{k=1}^m \frac{A_k}{(x-a)^k},$$

where the m constants A_k are to be determined. In the case of a nonrepeated quadratic factor $ax^2 + bx + c$ (where $b^2 - 4ac < 0$) we associate the single fraction

$$\frac{Ax + B}{ax^2 + bx + c},$$

where we now have two constants in the numerator to be determined, while if this quadratic factor is repeated m times, then we associate the sum

$$\sum_{k=1}^m \frac{A_k x + B_k}{(ax^2 + bx + c)^k},$$

which involves the determination of $2m$ constants A_k and B_k .

We illustrate the approach in the following example. Further discussion of partial fractions appears in Section 7.4, page 370.

Example 4 Find the indefinite integral $I_7 = \int \frac{3x^3 + 3x^2 + 3x + 2}{x^3(x+1)} dx$.

Solution Since the factor x is repeated three times in the denominator, while $(x+1)$ is a nonrepeated factor, our partial fraction decomposition must be

$$(11) \quad \frac{3x^3 + 3x^2 + 3x + 2}{x^3(x+1)} = \frac{A}{x+1} + \frac{B}{x} + \frac{C}{x^2} + \frac{D}{x^3},$$

where A , B , C , and D are unknowns to be determined. Multiplying both sides of (11) by $x^3(x+1)$, we have

$$(12) \quad 3x^3 + 3x^2 + 3x + 2 = Ax^3 + Bx^2(x+1) + Cx(x+1) + D(x+1).$$

In order for (12) to be true for all x , the corresponding coefficients must be the same:

$$\begin{array}{lll} \text{The coefficients of } x^3 \text{ yield} & A + B & = 3. \\ \text{The coefficients of } x^2 \text{ yield} & B + C & = 3. \\ \text{The coefficients of } x^1 \text{ yield} & C + D & = 3. \\ \text{The coefficients of } x^0 \text{ yield} & D & = 2. \end{array}$$

Solving this system of four linear equations in four unknowns gives $A = 1$, $B = 2$, $C = 1$, and $D = 2$. Hence

$$I_7 = \int \left(\frac{1}{x+1} + \frac{2}{x} + \frac{1}{x^2} + \frac{2}{x^3} \right) dx = \ln |(x+1)x^2| - \frac{1}{x} - \frac{1}{x^2} + C. \quad \blacklozenge$$

We remark that the coefficients A and D in the representation (11) can be determined directly. Indeed, if we set $x = 0$ in (12), we obtain $2 = D$, and if we set $x = -1$ in (12), we obtain $-3 + 3 - 3 + 2 = A(-1)$ or $A = 1$. But B and C cannot be obtained so readily. If we are fortunate to have a denominator with only nonrepeated linear factors, then this simple substitution method will work to quickly determine all the unknown coefficients.

In dealing with the integration of rational functions that have a quadratic factor in the denominator, the familiar formula

$$\int \frac{1}{1+x^2} dx = \arctan x + C$$

plays a central role. For example, by completing the square and making a simple change of variables, we find

$$\int \frac{3}{x^2 + 4x + 5} dx = \int \frac{3}{(x+2)^2 + 1} dx = 3 \arctan(x+2) + C.$$

A EXERCISES

In Problems 1–26, find the indicated indefinite integral.

1. $\int x^3(1+x^2)^{3/2} dx$

2. $\int x \cos(2x) dx$

5. $\int \cos^2(\pi\theta) d\theta$

6. $\int \frac{x^{1/3}}{x^{1/3} + 1} dx$

3. $\int 4t^4 e^{6t^5} dt$

4. $\int x^3 e^{3x} dx$

7. $\int \frac{t+1}{t^2+4} dt$

8. $\int 10 \sec^2(5x) dx$

9. $\int \frac{1}{\sqrt{1-2x^2}} dx$

10. $\int \frac{1}{2x^2 + 4x + 4} dx$

24. $\int \sin \sqrt{x} dx$

25. $\int \frac{\sin t}{1 + \cos^2 t} dt$

11. $\int \frac{4}{(x-1)(x-2)(x-3)} dx$

26. $\int \frac{e^{3y}}{1 + e^{6y}} dy$

12. $\int \sec(2x) \tan(2x) dx$

13. $\int \frac{\sqrt{9x^2 - 1}}{x} dx$

14. $\int \frac{2x + 41}{x^2 + 5x - 14} dx$

15. $\int \frac{x^3 + 2x^2 + 8}{x^2(x^2 + 4)} dx$

16. $\int \frac{x^3 + 4x + 1}{x^3 + 3x^2 + 4x + 2} dx$

17. $\int \frac{1}{x \ln x} dx$

18. $\int \frac{x^2}{x^2 - 1} dx$

19. $\int y \sinh y dy$

20. $\int \sec(3\theta + 1) d\theta$

21. $\int \frac{1}{(x^2 + 4)^{3/2}} dx$

22. $\int 36x^5 \exp(2x^3) dx$

23. $\int t \ln(t + 3) dt$

In Problems 27–34, use an appropriate trigonometric identity to help find the indicated integral.

27. $\int \sin^{2/5} x \cos^3 x dx$

28. $\int \tan^3 x \sqrt{\sec x} dx$

29. $\int \cos(3x) \cos(7x) dx$

30. $\int \cot^3 \theta d\theta$

31. $\int \tan^4 \theta \sec^6 \theta d\theta$

32. $\int \cos^{1/3} x \sin^3 x dx$

33. $\int \sin^2(3x) \cos^2(3x) dx$

34. $\int \cot^2 x \csc^4 x dx$